

Exchange and ventilation: two linked jobs

Teacher guide and worked answers

Explain how ventilation and alveolar structure support gas exchange, using an idealised comparison model.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C33

Explain exchange surfaces & ventilation.

C33 improved revisit of earlier diffusion/lung lessons, linked to Pearson 8.2-8.5B. Fick's-law reasoning is Biology-specific content, not labelled Higher-only. The core connects ventilation, gradients and short diffusion distance; no human breathing practical is required.

Prior learning

Diffusion is net movement down a concentration gradient.

Blood circulates between lungs and tissues.

A rate describes change per unit time.

Success evidence

Distinguish ventilation from diffusion and cellular respiration.

Explain thin, moist, large-area alveoli and blood flow using a mechanism.

Predict the effect of changing one diffusion factor while holding others constant.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The air moved. Has exchange happened yet?
2	Retrieval	2-5	What drives diffusion?
3	I do	5-9	Follow one gas at a time
4	I do	9-12	Air moves because pressure changes
5	I do	12-15	Explain the exchange adaptations
6	We do	15-18	Diffusion design desk
7	We do	18-20	Trace a ventilation failure in the model
8	We do	20-22	Repair a three-process caption
9	Hinge	22-25	Which explanation names the mechanism?
10	You do	25-26	Build an exchange explanation
11	You do	26-36	Make the link from structure to function
12	Review	36-38	Audience asks: what actually moves?
13	Review	38-39	Use three precise verbs
14	Exit	39-40	Three mechanism checks
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The air moved. Has exchange happened yet?

Invite pupils to distinguish opening a door from people moving through it as a limited analogy; return immediately to actual gas mechanisms.

2 Retrieval What drives diffusion?

R1: down a concentration gradient. R2: oxygen, with other transported substances. R3: no; cellular respiration releases energy in cells.

3 I do Follow one gas at a time

Think aloud along oxygen arrow, then carbon-dioxide arrow. State that gases cross the thin exchange surface by diffusion; blood and ventilation maintain differences. Do not claim all blood always has one uniform gas concentration.

4 I do Air moves because pressure changes

Model diaphragm contraction/flattening plus ribs up/out. Pressure falls below surrounding air pressure, so air enters. Quiet exhalation follows relaxation and elastic recoil; forced exhalation is beyond this core model.

5 I do Explain the exchange adaptations

Use the constructed model to illustrate one factor at a time. $P \rightarrow Q$ doubles area with other model factors equal; $P \rightarrow R$ doubles thickness. A relative index is not an oxygen saturation, clinical reading or empirical result.

6 We do Diffusion design desk

P/Q changes area only: Q is faster in the model. P/R changes thickness only: R is slower. Ask pupils to identify the unchanged columns before giving a conclusion.

7 We do Trace a ventilation failure in the model

Alveolar oxygen is removed into blood and not replenished as effectively; the difference can shrink, reducing net oxygen diffusion if other factors are comparable. This is a conceptual case; pupils do not hold their breath.

8 We do Repair a three-process caption

Diaphragm action supports ventilation; oxygen diffuses across alveoli into blood; cells use oxygen in aerobic respiration. No process has to be personified.

Reveal: Ventilation moves air. Diffusion exchanges gases. Cellular respiration releases energy in cells.

9 Hinge Which explanation names the mechanism?

First option correct. Use the two-panel diagrams to separate air movement from gas diffusion.

A. A thin exchange barrier helps because it shortens diffusion distance. - Yes, with other factors equal, a shorter path supports faster diffusion.

B. The diaphragm directly pushes oxygen through blood-cell membranes. - It changes thoracic volume/pressure; gas exchange across the lung barrier is diffusion.

C. A large surface always guarantees fast exchange, regardless of every other factor. - Gradient and distance also matter; one favourable factor is not a universal guarantee.

10 You do Build an exchange explanation

Core focuses on mechanisms and fair factor comparisons. The formula extension is Biology-specific reasoning, not inherently Higher-only.

11 You do Make the link from structure to function

Supported S1-S4; standard M1-M4; stretch T1-T4. Check the pressure direction before asking for extended prose.

12 Review Audience asks: what actually moves?

Pupil improves a process label or identifies a relevant gradient. Quiet teacher/TA audience is valid.

13 Review Use three precise verbs

Accept accurate wording rather than these exact verbs. Ask pupil to correct one misleading "pumps oxygen into cells" claim.

14 Exit Three mechanism checks

E1: falls below atmospheric pressure as volume rises. E2: shorter path. E3: ventilation/diffusion. No breathing exercise is requested.

15 Exit Next useful step

Shared exit minute; use an annotated diagram if writing demand is a barrier.

Answers and assessment

Retrieval and We do

R1: down a concentration gradient; R2: oxygen; R3: respiration differs from breathing. W1: P/Q changes area only; Q has twice the index. W2: P/R changes thickness only; R has half the index. W3: less fresh-air replacement can reduce alveolar oxygen and the oxygen gradient as exchange continues.

Caption repair and hinge

Ventilation moves air because volume/pressure changes. Oxygen diffuses into blood across alveoli. Cellular respiration releases energy in cells. Correct hinge: thin barrier shortens diffusion distance; diaphragm does not directly force oxygen across membranes.

S1 / M1

Oxygen arrow air → blood; carbon dioxide arrow blood → air under the stated gradients. Ventilation replenishes air and removes carbon dioxide; blood flow brings relatively oxygen-poor blood and carries exchanged gases away, helping maintain differences.

S2 / M3

During inhalation diaphragm contracts/flattens and external intercostals raise ribs up/out; thoracic volume rises, pressure falls below atmospheric and air enters. Quiet exhalation involves relaxation and elastic recoil, reducing volume and raising pressure so air leaves.

S3 / M4

P/Q differs only in area: Q faster. P/R differs only in thickness: R slower. Inputs are relative and idealised, so no clinical concentration, saturation or measured exchange rate can be read from the index.

S4 / M2

Large total area supports more simultaneous exchange; a thin alveolar/capillary barrier shortens the diffusion path; moisture enables gases to dissolve. Gradients and blood/air replacement are also important. Cellular respiration is an energy-releasing process in cells, not air movement.

T1

$P = 2 \times 4 / 1 = 8$; $Q = 4 \times 4 / 1 = 16$; $R = 2 \times 4 / 2 = 4$. $Q/P = 2$; $R/P = 0.5$. The numerator uses concentration difference; these are relative indices with the same proportionality constant.

T2

Unchanged: twice the area divided by twice the thickness gives the original index if the concentration difference and other model assumptions match.

T3

Small cube 6:1; larger cube 24:8=3:1. The larger cube has less surface per unit volume; increasing size can make diffusion-only exchange less adequate for its volume, depending on shape and demand.

T4

Idealised uniform barrier/gradient, no real units or diffusion coefficient, no changing blood flow/ventilation, no heterogeneous alveoli or individual physiology. Any two specific limits are valid.

Exit and assessment

E1: pressure falls as volume increases, so air moves in from higher external pressure. E2: shorter diffusion distance. E3: accurate ventilation/diffusion versus respiration distinction. Secure core requires a mechanism, not a feature list; reteach with one gas arrow and one pressure arrow.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

No pupil breath holding, forced breathing, lung-capacity testing or exercise is needed. The entire lesson uses models and supplied evidence.

Fick's-law factors and the optional relative-index calculation are Biology-specific in this specification; do not label them automatically Higher-tier.

Interactive and paper W1-W3 use identical model inputs. The animated arrows represent direction, not actual particle counts or an individual's gas exchange.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

Breathing, gas exchange and respiration are the same process.

Ventilation moves air; diffusion exchanges gases; cellular respiration releases energy in cells.

Check: Ask where each process occurs and what changes.

The lungs pump oxygen directly into every body cell.

Air enters by ventilation; oxygen diffuses into blood, which transports it towards tissues.

Check: Trace the separate air, exchange and blood stages.

A thicker barrier helps gases cross faster.

With other factors held constant, a longer diffusion distance reduces rate.

Check: Compare model P and R.

A relative model index is a real clinical gas reading.

It compares a specified model and has no clinical calibration.

Check: Ask what the index leaves out.

Word	Meaning
ventilation	Movement of air into and out of the lungs.
alveolus	A small air sac in the lungs where gases exchange with capillary blood.
concentration gradient	A difference in concentration across a distance.
diffusion distance	The distance particles must cross during exchange.
diaphragm	A muscular sheet that changes thoracic volume during breathing.
relative rate index	A model number for comparing rates; it is not a calibrated measurement of a person.

Scientific references

Pearson Edexcel GCSE Biology 1BIO specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

NHLBI: What breathing does for the body

Checked alveolar exchange and oxygen/carbon-dioxide transport. Checked 9 September 2026.

NHLBI: How the lungs work

Checked ventilation context. All diagrams and relative-index examples are original classroom models. Checked 9 September 2026.